Seminar 10

$[f]_E = [f(e_1) | f(e_2) | f(e_3)]$ the matrix of a linear map $[f]_E$.

$E = (e_1, e_2, e_3)$ basis of the vector space.

f

(domain)

$B = e \cdot S$ (basis)

$B' = e' \cdot T$ (basis)

(codomain)

$\Rightarrow [f]_{BB'} = T^{-1} \cdot [f]_{ee'} \, S$. Change of basis.

T $\downarrow$ matrix $B' \to e'$.

S $\downarrow$ matrix $B \to e$

$$[f]_E \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$ for a basis in $\operatorname{Ker}(f)$.

$\dim(\operatorname{Im} f) = \dim(f(e)) = \dim(\langle f(e_1), f(e_2), f(e_3), f(e_4) \rangle) =$
$=$ max. nr. of linearly indep. vectors in $[f]_E = \operatorname{rank}([f]_E)$.

f automorphism $\Leftrightarrow \begin{cases} \det(f[E]) \neq 0 \\ [2f]_E = 2[f]_E. \end{cases}$

① . $f \in \operatorname{End}_{\mathbb{R}}(\mathbb{R}^3)$, $f(x,y,z) = (x+y, y-z, 2x+y+z)$.

$[f]_E = ?$, $E = (e_1, e_2, e_3)$

$[f]_E = [f(e_1) | f(e_2) | f(e_3)]$

$$\begin{cases} f(e_1) = f(1,0,0) = (1+0, 0-0, 2+0+0) = (1,0,2) \\ f(e_2) = f(0,1,0) = (0+1, 1-0, 0+1+0) = (1,1,1) \\ f(e_3) = f(0,0,1) = \cancel{(0,0,2)} \, (0,-1,1). \end{cases}$$

$$\Rightarrow [f]_E = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & -1 \\ 2 & 1 & 1 \end{bmatrix}$$

(2.) $f \in Hom_{\mathbb{R}}(\mathbb{R}^3, \mathbb{R}^2)$, $f(x,y,z) = (y,-x)$.

$B = (v_1, v_2, v_3) = ((1,1,0),(0,1,1),(1,0,1)) \in \mathbb{R}^3$.

$B' = (v_1', v_2') = ((1,1),(1,-2)) \in \mathbb{R}^2$.

$E' = (e_1', e_2')$

$[f]_{BE'} = ?$

$[f]_{BB'} = ?$

$$\begin{cases} f(v_1) = f(1,1,0) = (1,-1) \\ f(v_2) = f(0,1,1) = (1,0) \\ f(v_3) = f(1,0,1) = (0,-1) \end{cases} \Rightarrow [f]_{BE'} = \begin{bmatrix} 1 & 1 & 0 \\ -1 & 0 & -1 \end{bmatrix}$$

$f(v_1) = (1,-1) \Rightarrow (1,-1) = a_1 \cdot v_1' + a_2 \cdot v_2' = \cancel{(a_1+a_2, a_1-a_2)}$.

$\Rightarrow (1,-1) = a_1 \cdot (1,1) + a_2 \cdot (1,-2)$

$\Rightarrow (1,-1) = (a_1+a_2, a_1-2a_2)$

$$\Rightarrow \begin{cases} a_1 + a_2 = 1 \\ a_1 - 2a_2 = -1 \end{cases} \quad (-)$$

$3a_2 = 2 \Rightarrow a_2 = \frac{2}{3}$, $a_1 = 1 - \frac{2}{3} = \frac{1}{3}$

$f(v_2) = (1,0) \Rightarrow (1,0) = a_1 v_1' + a_2 v_2'$

$\Rightarrow (1,0) = a_1 (1,1) + a_2 (1,-2)$

$\Rightarrow (1,0) = (a_1 + a_2, a_1 - 2a_2)$

$$\Rightarrow \begin{cases} a_1 + a_2 = 1 \\ a_1 - 2a_2 = 0 \end{cases} \quad (-)$$

$3a_2 = 1 \Rightarrow a_2 = \frac{1}{3}$, $a_1 = 1 - \frac{1}{3} = \frac{2}{3}$.

$f(v_3) = (0,-1) \Rightarrow (0,-1) = a_1(1,1) + a_2(1,-2)$

$(0,-1) = (a_1 + a_2, a_1 - 2a_2)$

$$\begin{cases} a_1 + a_2 = 0 \\ a_1 - 2a_2 = -1 \end{cases} \quad (-)$$

$3a_2 = 1 \Rightarrow a_2 = \frac{1}{3}$, $a_1 = -\frac{1}{3}$

$$[f]_{BB'} = \begin{bmatrix} \frac{1}{3} & \frac{2}{3} & -\frac{1}{3} \\ \frac{2}{3} & \frac{1}{3} & \frac{1}{3} \end{bmatrix}$$

$\times$ $f \in Hom_{\mathbb{R}}(\mathbb{R}^3, \mathbb{R}^4)$

$f(e_1) = (1,2,3,4)$

$f(e_2) = (4,3,2,1)$

$f(e_3) = (-2,1,4,1)$

a) $f(v)$, $(\forall) v \in \mathbb{R}^3 \Rightarrow v = (x,y,z) \in \mathbb{R}^3$.

$f(v) = (x \cdot e_1 + y e_2 + z \cdot e_3) \overset{f \text{ hom.}}{=\!=\!=} x \cdot f(e_1) + y \cdot f(e_2) + z \cdot f(e_3)$.

$\Rightarrow f(v) = x \cdot (1,2,3,4) + y \cdot (4,3,2,1) + z \cdot (-2,1,4,1)$

$= (\underline{x+4y-2z}, \underline{2x+3y+z}, \underline{3x+2y+4z}, \underline{4x+y+z}) \in \mathbb{R}^4$.

b) $[f]_E = \begin{bmatrix} 1 & 4 & -2 \\ 2 & 3 & 1 \\ 3 & 2 & 4 \\ 4 & 1 & 1 \end{bmatrix}$?

$4\times\cancel{3}, \cancel{3}\times 1 = 4\times 1$. ✓

$\underline{4\times\cancel{3} \cdot \cancel{3}\times 1}$

c) $[f]_E \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{cases} x_1 + 4x_2 - 2x_3 = 0 \\ 2x_1 + 3x_2 + x_3 = 0 \\ 3x_1 + 2x_2 + 4x_3 = 0 \\ 4x_1 + x_2 + x_3 = 0 \end{cases}$.

$\boxed{Ax = 0} \rightarrow$ all sols $\downarrow$ Ker.

o) $A = \begin{bmatrix} 1 & 4 & -2 \\ 2 & 3 & 1 \\ 3 & 2 & 4 \\ 4 & 1 & 1 \end{bmatrix}$, $\bar{A} = \left[\begin{array}{ccc|c} 1 & 4 & -2 & 0 \\ 2 & 3 & 1 & 0 \\ 3 & 2 & 4 & 0 \\ 4 & 1 & 1 & 0 \end{array}\right] \xrightarrow{L_4 \leftarrow L_4 - L_2}$

$= \left[\begin{array}{ccc|c} 1 & 4 & -2 & 0 \\ 2 & 3 & 1 & 0 \\ 3 & 2 & 2 & 0 \\ 2 & -2 & 0 & 0 \end{array}\right] \xrightarrow{L_3 \leftarrow L_3 + L_4} \left[\begin{array}{ccc|c} 1 & 4 & -2 & 0 \\ 2 & 3 & 1 & 0 \\ 5 & 0 & 2 & 0 \\ 2 & -2 & 0 & 0 \end{array}\right] \xrightarrow{L_2 \leftarrow L_2 - L_4} \left[\begin{array}{ccc|c} 1 & 4 & -2 & 0 \\ 0 & 5 & 1 & 0 \\ 5 & 0 & 2 & 0 \\ 2 & -2 & 0 & 0 \end{array}\right]$

$\left[\begin{array}{ccc|c} 1 & 4 & -2 & 0 \\ 2 & 3 & 1 & 0 \\ 5 & 5 & 5 & 0 \\ 4 & 1 & 1 & 0 \end{array}\right] \xrightarrow{L_2 \leftarrow L_2 - L_4} \left[\begin{array}{ccc|c} 1 & 4 & -2 & 0 \\ -2 & 2 & 0 & 0 \\ 5 & 5 & 5 & 0 \\ 4 & 1 & 1 & 0 \end{array}\right]$

3 2;

3 2 4

-2 2 2

$\xrightarrow{L_4 \leftarrow L_4 - 2L_2} \left[\begin{array}{ccc|c} 1 & 4 & -2 & 0 \\ 2 & 3 & 1 & 0 \\ 3 & 2 & 4 & 0 \\ 0 & -5 & -1 & 0 \end{array}\right] = \left[\begin{array}{ccc|c} 1 & 4 & -2 & 0 \\ 2 & 3 & 1 & 0 \\ 5 & 5 & 5 & 0 \\ 0 & -5 & -1 & 0 \end{array}\right] =\!=\!=$

1-6

$$\left[\begin{array}{ccc|c} 1 & 4 & -2 & 0 \\ 2 & 3 & 1 & 0 \\ 3 & 2 & 4 & 0 \\ 4 & 1 & 1 & 0 \end{array}\right] \xrightarrow{R_2 \leftarrow R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 4 & -2 & 0 \\ 0 & -5 & 5 & 0 \\ 3 & 2 & 4 & 0 \\ 4 & 1 & 1 & 0 \end{array}\right] \xrightarrow{R_3 \leftarrow R_3 - 3R_1} \left[\begin{array}{ccc|c} 1 & 4 & -2 & 0 \\ 0 & -5 & 5 & 0 \\ 0 & -10 & 10 & 0 \\ 4 & 1 & 1 & 0 \end{array}\right] =$$

$$\xrightarrow{R_4 \leftarrow R_4 - 4R_1} \left[\begin{array}{ccc|c} 1 & 4 & -2 & 0 \\ 0 & -5 & 5 & 0 \\ 0 & -10 & 10 & 0 \\ 0 & -15 & 9 & 0 \end{array}\right] \xrightarrow[\substack{R_3 \leftarrow -\frac{1}{10} R_3 \\ R_4 \leftarrow -\frac{1}{3} R_4}]{R_2 \leftarrow -\frac{1}{5} R_2} \left[\begin{array}{ccc|c} 1 & 4 & -2 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 3 & -3 & 0 \end{array}\right] \xrightarrow{R_3 \leftarrow R_3 - 3R}$$

$$= \left[\begin{array}{ccc|c} 1 & 4 & -2 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{R_3 \leftarrow R_3 - R_2} \left[\begin{array}{ccc|c} 1 & 4 & -2 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\Rightarrow \begin{cases} x + 4y - 2z = 0 \\ \cancel{x} + y - z = 0 \end{cases} \quad (-)$$

$$3y - z = 0 \Rightarrow 3y = z.$$

$$\Rightarrow \begin{cases} x + 4y - 2z = 0. \\ x + y - 3y = 0 \Rightarrow x - 2y = 0 \Rightarrow x = 2y. \end{cases}$$

$$\cancel{x} \quad 2y + 4y - 2\cdot(3y) = 0$$

$$6y - 6y = 0$$

$$\begin{cases} x + 4y - 2z = 0 \\ y - z = 0 \Rightarrow y = z \end{cases} \Rightarrow x + 4y - 2y = 0$$

$$\Rightarrow x + 2y = 0$$

$$\Rightarrow \begin{cases} x = -2y = -2\alpha \\ z = y = \alpha \\ y = y = \alpha \end{cases} =$$

$$\Rightarrow \operatorname{Ker} f = \langle (-2,1,1) \rangle,$$

$$\Rightarrow \dim(\operatorname{Ker} f) = 1.$$

$\operatorname{Im} f$:

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ -2 & 1 & 4 & 1 \end{bmatrix} \xrightarrow{R_2 \leftarrow R_2 - 4R_1} \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & -5 & -10 & -15 \\ -2 & 1 & 4 & 1 \end{bmatrix} \xrightarrow{R_2 \leftarrow -\frac{1}{5} R_2}$$

$$= \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ -2 & 1 & 4 & 1 \end{bmatrix} \xrightarrow{R_3 \leftarrow R_3 + 2R_1} \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 5 & 10 & 9 \end{bmatrix} \xrightarrow{R_3 \leftarrow R_3 - \cancel{4}R_2}$$

Side calculations:

$2 - 12$
-10

$1.$
$1 - 4\cdot(-2)$
$\cancel{1-4}$ $1 + 8$

$1 - 3\cdot(-2)$
$1 + 6$

$4 - 2\cdot(-3)$
$4 + 6$

$\cancel{-2z+z}$

$4 \;\; 8 \;\; 6$
$2 \;\; 4 \;\; 3$

$5 - 2$
$10 - 4$

$3 - 8$
$2 - 4\cdot 3$
$2 - 12$

$\cancel{1}$ $1 +$
$\cancel{2}$ $4 + 6$

$$= \begin{bmatrix}1&2&3&4\\0&1&2&3\\0&4&8&6\end{bmatrix} \xrightarrow{R_3 \leftarrow \frac{1}{2}R_3} \begin{bmatrix}1&2&3&4\\0&1&2&3\\0&2&4&3\end{bmatrix} \xrightarrow{R_3 \leftarrow R_3 - 2R_2} \begin{bmatrix}1&2&3&4\\0&1&2&3\\0&0&0&-3\end{bmatrix}$$

$\Rightarrow$ ~~dim~~ $\operatorname{Im} f = \langle (1,2,3,4), (0,1,2,3), (0,0,0,-3) \rangle$

$\Rightarrow \dim \operatorname{Im} f = 3 = \underline{\text{rank}}$.

4. $f \in \operatorname{End}_{\mathbb{R}}(\mathbb{R}^4)$.

$$[f]_E = \begin{bmatrix}1&1&-3&2\\-1&1&1&4\\2&1&-5&1\\1&2&-4&5\end{bmatrix}$$

a) $v = (1,4,1,-1) \overset{?}{\in} \operatorname{Ker} f$

$v' = (2,-2,4,2) \overset{?}{\in} \operatorname{Im} f$

$v = (1,4,1,-1) = e_1 + 4e_2 + e_3 - e_4 \mid f$

$\Rightarrow f(v) = f(e_1) + 4f(e_2) + f(e_3) - f(e_4)$

$\Rightarrow f(v) = (1,-1,2,1) + 4(1,1,1,2) + (-3,1,-5,-4) - (2,4,1,5)$

$\Rightarrow f(v) = (1+4-3-2, -1+4+1-4, 2+4-5-1, 1+8-4-5)$

$= \underline{(0,0,0,0)} \Rightarrow v \in \operatorname{Ker} f$.

$\boxed{v' \in \operatorname{Im} f \Leftrightarrow \exists v \text{ s.t. } f(v) = v'}$

$v' = a \cdot f(e_1) + b \cdot f(e_2) + c \cdot f(e_3) + d \cdot f(e_4)$

$(2,-2,4,2) = a(1,-1,2,1) + b \cdot (1,1,1,2) + c(-3,1,-5,-4) + d(2,4,1,5)$

$= (a+b-3c+2d, -a+b+c+4d, 2a+b-5c+d, a+2b-4c+5d)$

$$\Rightarrow \begin{cases} a+b-3c+2d = 2 \\ -a+b+c+4d = -2 \\ 2a+b-5c+d = 4 \\ a+2b-4c+5d = 2 \end{cases}$$

$(+) \Rightarrow 2b - 2c + 6d = 0 \Rightarrow \underline{b - c + 3d = 0}$

$\underline{b = c - 3d.}$

$(2) + (4) \Rightarrow 3b - 3c + 9d = 0 \mid :3 \Rightarrow b - c + 3d = 0$

$2a + c - 3d - 5c + d = 4 \Rightarrow 2a - 4c - 2d = 4. \mid :2 \Rightarrow \underline{a - 2c - d = 2}$

$$\begin{cases} \underline{a = 2c + d + 2} \\ b = c - 3d \end{cases}$$

$2c + d + \cancel{2} + 2(c - 3d) - 4c + 5d = \cancel{2}$

$\cancel{4c} - \cancel{4c} + d - 6d + 5d = 0 \Rightarrow d - d = 0.$

$\cancel{2c} + d + \cancel{2} + \cancel{c} - 3d - \cancel{3c} + 2d = \cancel{2} \Rightarrow$

$$[f]_E = \begin{bmatrix} 1 & 1 & -3 & 2 \\ -1 & 1 & 1 & 4 \\ 2 & 1 & -5 & 1 \\ 1 & 2 & -4 & 5 \end{bmatrix}$$

$$\boxed{\dim \operatorname{Im} f = \dim \langle f(e) \rangle}$$

$$\boxed{= \operatorname{rank} [f]_E} \; = 2 \; ??$$

$\Rightarrow \dim \operatorname{Im} f = 2$

$\operatorname{Im} f = \langle (1,1,-3,2), (-1,1,1,4) \rangle$

$$-\varepsilon < x_n - l < \varepsilon \quad | + l$$

$$l - \varepsilon < x_n < l + \varepsilon$$

$\cancel{2n - 1 - 2n - 1}$

$-1 + (-1)^2 + (-1)$

$\cancel{-1 + 1 - 1 + 1}$

$-1 + 1 - 1$

$$\boxed{\dim V = \dim \ker f + \dim \operatorname{Im} f}$$

$4 = 2 + \dim \operatorname{Im} f$

$\dim \operatorname{Im} f = 2.$

$-1 \quad 1 \quad 0$

$-4 + 5$

$-1 \quad 1 \quad 1 \quad 4$

$\underline{R_4 \leftarrow R_4 - R_3}$
$$\begin{bmatrix} 1 & 1 & -3 & 2 \\ -1 & 1 & 1 & 4 \\ 2 & 1 & -5 & 1 \\ -1 & 1 & 1 & 4 \end{bmatrix} \xrightarrow{R_4 \leftarrow R_4 - R_2} \begin{bmatrix} 1 & 1 & -3 & 2 \\ -1 & 1 & 1 & 4 \\ 2 & 1 & -5 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$\underline{R_2 \leftarrow R_2 - R_1}$
$$\begin{bmatrix} 1 & 1 & -3 & 2 \\ -2 & 0 & 4 & 2 \\ 2 & 1 & -5 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_3 \leftarrow R_3 - R_1} \begin{bmatrix} 1 & 1 & -3 & 2 \\ -2 & 0 & 4 & 2 \\ 1 & 0 & -2 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$\underline{R_2 \leftarrow R_2 + 2R_3}$
$$\begin{bmatrix} 1 & 1 & -3 & 2 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & -2 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{bmatrix} 1 & 1 & -3 & 2 \\ 1 & 0 & -2 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$-5 + 3$

$4 -$

$2 - 2$

-2

-1

$\underline{R_2 \leftarrow R_2 - R_1}$
$$\begin{bmatrix} 1 & 1 & -3 & 2 \\ 0 & -1 & 1 & -3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$\Rightarrow \operatorname{rank} [f]_E = 2.$

~~c = b + 3d.~~

~~a = 2(b + 3d) + d + 2~~

$$\begin{cases} 2a+b-5c+d=4 \\ b-c+3d=0 \\ a-2c-d=2 \mid \cdot 3 \end{cases} \Rightarrow \begin{cases} b-c+3d=0 \\ 3a-6c-3d=6 \end{cases} \quad (+)$$

$\Downarrow$

~~a-2c=2~~

$d = a-2c-2.$

$$\Rightarrow \begin{cases} 3a+b-7c=6 \\ 2a+b-5c+d=4. \end{cases}$$

$2a+b-5c+a-2c-2=4.$

$3a+b-7c=6.$

? $\quad (-)$

$a-2c+d=2$

$\Rightarrow \boxed{b = c-3d}$

$\boxed{a = 2+2c+d}$

$c, d \in \mathbb{R}$ (termenii liberi)

$\to$ sol. $(2+2c+d,\ c-3d,\ c,\ d)$.

$\Rightarrow \exists v$ s.t. $f(v) = v'$.

$\Rightarrow$ ~~v~~ de sol (a, b, c, d)

b) basis, dim of Ker f, Im f.

$$[f]_E \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \to \begin{bmatrix} 1 & 1 & -3 & 2 \\ -1 & 1 & 1 & 4 \\ 2 & 1 & -5 & 1 \\ 1 & 2 & -4 & 5 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow$$

$$\Rightarrow \begin{cases} x_1+x_2-3x_3+2x_4=0 \\ -x_1+x_2+x_3+4x_4=0 \\ 2x_1+x_2-5x_3+x_4=0 \\ x_1+2x_2-4x_3+5x_4=0. \end{cases}$$

$\left.\begin{array}{l} \dim \operatorname{Im} f = \operatorname{rank} [f]_E \\ \operatorname{rank} [f]_E = 2 \end{array}\right\} \Rightarrow \dim \operatorname{Im} f = 2$

$\Rightarrow \operatorname{Im} f = \langle (1,1,-3,2), (-1,1,1,4) \rangle$

$x_2 = x_3 - 3x_4$

$x_1 = 2x_3 + x_4 + \cancel{2}$

$x_3 = \cancel{a}\ b$

$x_4 = \cancel{b}\ a$

ex: $x_3 = 1$

$x_4 = 1$

$x_2 = 1-3 = -2$

$x_1 = 2 \cdot 1 + 1 + 2 = 5$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 2b+a \\ b-3a \\ b \\ a \end{bmatrix} = a \cdot \begin{bmatrix} 1 \\ -3 \\ 0 \\ 1 \end{bmatrix} + b \cdot \begin{bmatrix} 2 \\ 1 \\ 1 \\ 0 \end{bmatrix} \Rightarrow \operatorname{Ker} f = \langle (1,-3,0,1), (2,1,1,0) \rangle$$

$\dim \operatorname{Ker} f = 2.$

c) define f.

$$\begin{cases} f(1,0,0,0) = (1,-1,2,1) = (x,-x,2x,x) \\ f(0,1,0,0) = (1,1,1,2) = (y,y,y,2y) \\ f(0,0,1,0) = (-3,1,-5,-4) = (-3z, z, -5z, -4z) \\ f(0,0,0,1) = (2,4,1,5) = (2t,4t,t,5t) \end{cases}$$

$\Rightarrow f(x,y,z,t) = (x+y-3z+2t,\ -x+y+z+4t,\ 2x+y-5z+t,\ x+2y-4z+5t)$

♡ ⑤. $\mathbb{R}_2[X] = \{ f \in \mathbb{R}[X] \mid \text{degree}(f) \leq 2 \}$, $E = (1, x, x^2)$

$B = (\overset{b_1}{1}, \overset{b_2}{x-1}, \overset{b_3}{x^2+1})$

$\varphi \in \text{End}_{\mathbb{R}}(\mathbb{R}_2[X])$: $\varphi(a_0 + a_1 x + a_2 x^2) = (a_0 + a_1) + (a_1 + a_2)x + (a_0 + a_2)x^2$

$[\varphi]_E = ?$, $[\varphi]_B = ?$

$\varphi(e_1) = \varphi(\underline{1} \cdot 1 + \underline{0} \cdot x + \underline{0} \cdot x^2) = (1+0) + (0+0)x + (1+0)x^2 = \underline{1 + x^2}$

$\varphi(e_2) = \varphi(\underline{0} \cdot 1 + \underline{1} \cdot x + \underline{0} \cdot x^2) = (0+1) + (1+0)x + (0+0)x^2 = \underline{1 + x}$

$\varphi(e_3) = \varphi(\underline{0} \cdot 1 + \underline{0} \cdot x + \underline{1} \cdot x^2) = (0+0) + (0+1)x + (0+1)x^2 = \underline{x + x^2}$

$$[\varphi]_E = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}$$

$\cancel{1} + x\ \cancel{1} + x^2 + 1$

$1 + x + x^2$,

$\varphi(b_1) = \varphi(1 \cdot 1 + 0 \cdot (x-1) + 0 \cdot (x^2+1)) =$

$\cancel{(1,0,0) \to (1,}$

$\varphi(b_1) = \varphi(1) = (1 + 0 \cdot x + 0 \cdot x^2) = (1+0) + (0+0)x + (1+0)x^2 = \underline{1 + x^2}$.

$\varphi(b_2) = \varphi(x-1) = ((-1) + 1 \cdot x + 0 \cdot x^2) = (-1+1) + (+1+0)x + (-1+0)x^2 = \underline{+x - x^2}$

$\varphi(b_3) = \varphi(x^2+1) = (1 + 0 \cdot x + 1 \cdot x^2) = (1+0) + (0+1)x + (1+1)x^2 = \underline{x + 2x^2 + 1}$

$$[\varphi]_B = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & -1 & 2 \end{bmatrix}$$

⑥. $B = (v_1, v_2) = ((1,2),(1,3))$

$B' = (v_1', v_2') = ((1,0),(2,1))$ $\Rightarrow$ $\det \neq 0 \Rightarrow$ automorph.

$f, g \in \text{End}_{\mathbb{R}}(\mathbb{R}^2)$, $[f]_B = \begin{pmatrix} 1 & 2 \\ -1 & -1 \end{pmatrix}$, $[g]_{B'} = \begin{pmatrix} -7 & -13 \\ 5 & 7 \end{pmatrix}$

$[2f]_B = ?$ $|[f]_B| = \begin{vmatrix} 1 & 2 \\ -1 & -1 \end{vmatrix} = -1 + 2 = 1 \neq 0 \Rightarrow f$ automorphism. $\Rightarrow$

$\Rightarrow \underline{[2f]_B = 2[f]_B} = 2 \cdot \begin{pmatrix} 1 & 2 \\ -1 & -1 \end{pmatrix} = \begin{pmatrix} 2 & 4 \\ -2 & -2 \end{pmatrix} \Rightarrow \begin{cases} f(v_1) = (2,-2) = (1,-1) \\ f(v_2) = (4,-2) = (2,-1) \end{cases}$

$\Rightarrow \begin{cases} a_1 \cdot x + b_1 \cdot y \overset{?}{=} 1 \\ a_2 x + b_2 y = -1 \end{cases}$; $v_1 = (1,2) \Rightarrow x = 1, y = 2 \Rightarrow \begin{cases} a_1 + 2b_1 = 1 \Rightarrow a_1 = -1 \\ a_2 + 2b_2 = -1 \quad a_2 = -1 \end{cases}$

$v_2 = (1,3) \Rightarrow \begin{cases} a_1 + 3b_1 = 2 \\ a_2 + 3b_2 = -1 \end{cases} \Rightarrow b_1 = +1,\ b_2 = +0$

$f(x,y) = (y-x, -x)$ ...

⑦ $f: \mathbb{R}^2 \to \mathbb{R}^2,\ f(x,y) = (x\cos\alpha - y\sin\alpha,\ x\sin\alpha + y\cos\alpha)$

$[f]_E = \begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix}$ $\quad f(e_1) = f(1,0) = (\cos\alpha - 0,\ \sin\alpha + 0) = (\cos\alpha, \sin\alpha)$

$f(e_2) = f(0,1) = (0 - \sin\alpha,\ \cos\alpha) = (-\sin\alpha, \cos\alpha)$

$\det [f]_E = \begin{vmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{vmatrix} = \cos^2\alpha + \sin^2\alpha = 1 \neq 0 \Rightarrow f$ automorphism.

⑧ $\dim_{\mathbb{Z}_2}$ ...

$2[f]_E \overset{?}{=} [2f]_E$?

$2\begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix} =$